

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY FOURTH SEMESTER B.TECH DEGREE EXAMINATION, MAY 2021		
Course Code: EST200		
Course Name: Design and Engineering		
Max. Marks: 100		Duration: 3 Hours
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
	1	Understanding customer requirements helps • to communicate and define customer needs and problems • to estimate project expenditure involved to set timelines
	2	Constraints that can occur in design process of a lunch box include but not limited to • Economic factors • Safety • Reliability • Aesthetics
	3	Convergent thinking is selecting the optimal solution from a finite set of ideas collected from different sources in order to solve a discrete challenge quickly and efficiently.
	4	Conflict, in the design process, helps validate and elaborate the ideas on which the whole process is established. It adds new perspectives and improves productivity.
	5	One of the best ways to gain insights in a Design Thinking process is to carry out some form of prototyping. Prototyping offers designers the opportunity to bring their ideas to life, test the practicability of the current design, and to potentially investigate how a sample of users think and feel about a product.
	6	Writing is a permanent means of communication. It is useful where record maintenance is required. It assists in proper delegation of responsibilities. Written communication is more precise and explicit. It provides ready records

		and reference.	
	7	Reverse engineering is cost effective. It can reproduce products and parts without needing the original design measurements and specifications. Reverse engineering can save time and human resources.	
	8	Bio mimicry is a technological-oriented approach focused on putting nature's lessons into practice. Bio mimicry studies nature's models and imitates them or uses them as inspiration for designs or processes with the goal of solving human problems. (2 mark) Example: (1 mark)	
	9	Engineering design ethics concerns issues that arise during the design of technological products, processes, systems, and services. This includes issues such as safety, sustainability, user autonomy, and privacy. (any 2 ethics-3 marks)	
	10	Sustainability in design • Ensures a healthy habitat • Reduce energy usage Ensures a future for all	
PART B			
<i>Answer any one full question from each module. Each question carries 14 marks</i>			
		Module 1	
	11	Explain the design process through different stages including Customer Requirements, Setting Design Objectives, Identifying Constraints, Establishing Functions, Generating Design Alternatives & Choosing a Design. Explanation(5), identifying the need of defined persona and design(5), Sketches(4)	
	12	Brief note of design process(3) • Engineering design is the process of devising a system, component, or process to meet desired needs • decision-making process (often iterative), in which the basic sciences, mathematics, and engineering sciences are applied to optimally convert resources to meet a stated objective • Fundamental elements of design process- ✓ establishment of objectives and criteria ✓ Synthesis ✓ Analysis ✓ Construction, testing, and evaluation detailed explanation of design alternatives and designs of coffee mug (7), Sketches(4)	
		Module 2	

	13	Explanation of design thinking process including Empathize, Define, Ideate, Prototype and Test(6), design of walking stick (4), Sketches(4) • Empathise- work to fully understand experience of the user for whom we are designing; done with observation, interaction, and immersing in their experiences • Define – process and synthesize findings from your empathy work to form a user point of view that will be addressed by your design • Ideate – explore wide variety of possible solutions through generating large quantity of possible solutions, allowing stepping beyond the obvious, exploring a range of ideas • Prototype – transform ideas into physical form • Test- try out high resolution products, use observations and feedbacks to refine prototypes
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	14	Brief note of design thinking process(5), design of parachute with iterative design process(5) sketches(4)
		Module 3
	15	Notes on communication of designs through sketches (graphical communication)(5), design of office chair through sketches(5) notes on material selection, scale drawings and dimensions(4) • Sketches- generated during conceptualisation stage of design process • Different mechanisms of each function and subfunctions are sketched • Sketches are then used to build devices • Working documents that support development of major components and their relations • Usually made to scale- but only with important dimensions shown in layout
	16	Explanation of mathematical modelling in design(7), Suitable example to explain the role of mathematical modelling(7) • Most engineering designs- complex in nature • Simplify design, reduce the number of parts because for each part, there is an opportunity for a defective part and an assembly error • Probability of a perfect product goes down exponentially as the number of parts increases • As number of parts goes up, total cost of fabricating and assembling the product goes up • Designer should go through the assembly part by part and evaluate whether the part can be eliminated, combined with another part, or the function can be performed in another way
		Module 4
	17	Explanation of modular approach(7), Explaining modular approach through design of desktop computers of different configuration(7) ▪ Module' means separate elements ▪ Modular design is an approach in which a product is designed for assembling in module-wise fashion ▪ Modular products are the artifacts that are composed of many modules ▪ These modules function together to get the overall function of the product ▪ Modular products can be machines, assemblies and components that fulfill various overall functions through the combination of distinct building blocks or modules ▪ In a modular product (or modular system), the overall function performed by the product is the results achieved through a combination of discrete units (modules) ▪ Dividing a product into discrete units based on some criteria is called as modularization of a product ▪ As we have seen, modular products or modular Systems are built up on separable or inseparable units called as modules ▪ The basic idea behind modular design is to organize a complex system as a set of distinct component that can be developed independently and then assembled together to perform a function

18	Explanation of ergonomics and its importance in engineering design (7) design of table lamp with ergonomics concept (usability to all) ▪ Ergonomic design applied to an office chair will focus on how much it is comfortable for a Person who sits on it during office work ▪ A chair ergonomically designed for dining purpose and a chair meant for relaxed sitting at beach will be different ▪ It is, because, the kind of comfort and function to be provided by the chairs in these situations are different ▪ To develop an ergonomic design (for a product or system), the designer will have to consider and analyse anthropometric data (dimensions of human body), posture of working while using the product, kind of movements and kind of workspace ▪ ultimately, ergonomic design involves every aspect of user-product interaction, for the comfortable utilization of a product
	Module 5
19	Explanation of design process of tea cup(7) Explanations of changes in design with concepts of production, use and sustainability(7) ✓ Design for manufacturing (DFM) is design based on minimizing the costs of production and/or the time to market for a product, while maintaining an appropriate level of quality. ✓ DFM begins with the formation of the design team. ✓ In commercial settings, design teams committed to DFM tend to be multidisciplinary, and they include engineers, manufacturing managers, logistics specialists, cost accountants, and marketing and sales professionals. ✓ Each brings particular interests and experience to a design project, but all must move beyond their primary expertise to focus on the project itself ✓ Reliability ✓ To an engineer, reliability is defined as “the probability that an item will perform its function under stated conditions of use and maintenance for a stated measure of a variate ✓ we can properly measure the reliability of a component or system only under the assumption that it has been or will be used under some specified conditions. ✓ the appropriate measure of use of the design, called the variate, may be something other than time. ✓ The American Society of Mechanical Engineers (ASME) stipulates that “Engineers shall consider environmental impact in the performance of their duties.” ✓ Environmental Issues and Design ✓ We can often characterize the environmental implications of a design in terms of the effects on air quality, water quality, energy consumption, and waste generation ✓ Air quality almost immediately springs to mind when we list environmental concerns related to design. ✓ Environmentally conscious engineers should also concern themselves with issues of water quality and water consumption
20	Explanation of cost estimation of table with labour, materials and overhead costs(10) Notes on influence of economics in design(4) • There is an important distinction between the cost of designing, prototyping, and testing a product and the cost of the product after manufacturing and distribution • while costing is an important element in the profitability of a design, it is generally not a key factor in the pricing of the artifact • costs are an important element in the profit equation. Revenues, on the other hand,

		are determined by the price charged for an item multiplied by the number of items sold • For most profit maximizing firms, prices are not set on the basis of costs, but rather in terms of what the market is willing to pay • the responsibilities of marketing professionals on a design team usually include identifying design attributes that make consumers willing to pay a high price for a new product design
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